

JYVASKYLA, Finland

WMO: 029350

Lat: 62.398N

Lon: 25.671E

Elev: 459

StdP: 14.45

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.2	-11.2	-22.2	1.6	-17.1	-15.6	2.4	-11.1	22.2	26.0	19.0	27.3	2.6	320	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	80.1	63.9	76.6	62.5	73.4	60.9	66.9	75.3	64.8	72.5	62.9	70.3	7.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	90.8	70.0	62.1	84.8	68.1	59.9	78.2	65.9	31.6	75.1	30.1	72.5	28.6	70.0	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.5	13.5	DB	-22.3	84.0	7.6	3.5	-27.8	86.6	-32.3	88.6	-36.5	90.6	-42.1	93.2	(4)
(5)			WB	-22.6	69.1	7.5	2.9	-28.0	71.3	-32.4	73.0	-36.6	74.6	-42.1	76.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.4	17.3	17.4	24.6	36.3	47.3	56.1	61.7	58.2	48.8	38.5	30.5	23.0		
	DBStd	17.72	13.07	13.38	9.39	6.94	7.85	6.40	5.75	5.63	6.17	7.38	8.44	12.20		
	HDD50	5167	1012	913	786	412	145	15	1	5	95	361	586	836		
	HDD65	9748	1477	1333	1251	860	548	275	138	221	485	823	1036	1301		
	CDD50	948	0	0	0	2	63	199	362	259	60	3	0	0		
	CDD65	55	0	0	0	0	1	9	34	11	0	0	0	0		
	CDH74	592	0	0	0	0	36	116	334	106	0	0	0	0		
	CDH80	87	0	0	0	0	3	13	60	11	0	0	0	0		
	WSAvg	6.0	6.0	6.1	6.5	5.9	6.4	6.2	5.6	5.1	5.5	5.9	6.3	6.3		
	PrecAvg	24.8	1.7	1.2	1.3	1.4	1.7	2.4	3.1	3.2	2.4	2.4	2.2	1.9		
(15) Precipitation	PrecMax	33.3	3.2	3.3	2.8	3.1	4.5	6.2	5.9	7.2	5.0	5.1	4.5	3.7		
	PrecMin	15.6	0.5	0.1	0.1	0.1	0.2	0.3	0.2	0.6	0.5	0.6	0.2	0.5		
	PrecStd	3.7	0.7	0.7	0.6	0.7	0.9	1.1	1.5	1.6	1.0	1.1	0.9	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	40.7	49.4	65.6	78.7	82.0	84.9	81.0	69.6	56.7	49.0	44.0		
		MCWB	37.5	37.0	40.4	50.3	59.2	63.9	66.5	64.8	59.8	52.1	47.7	42.8		
	2%	DB	36.0	37.3	43.2	58.4	73.0	77.4	81.4	76.9	65.8	53.1	45.9	40.8		
		MCWB	35.1	35.8	36.8	47.0	56.2	61.1	65.6	63.4	57.8	51.0	44.8	39.9		
	5%	DB	34.3	34.8	39.6	53.5	67.8	73.3	77.6	73.7	62.3	51.4	43.2	38.0		
		MCWB	33.6	33.9	34.9	43.8	53.3	59.4	63.8	61.6	56.0	49.6	42.3	37.1		
	10%	DB	32.8	33.0	36.9	49.0	63.0	69.6	74.2	70.2	59.3	49.3	41.0	35.9		
		MCWB	32.3	32.3	33.7	41.6	51.0	57.5	62.5	60.5	54.6	47.7	40.1	35.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.7	37.8	41.4	53.2	61.9	66.8	70.6	68.3	61.2	53.8	47.9	42.9		
		MCDB	38.1	39.2	46.9	60.8	72.5	76.4	79.6	77.3	66.5	55.5	48.8	43.4		
	2%	WB	35.2	35.9	38.0	48.3	58.5	63.8	67.9	65.4	59.1	51.7	44.6	39.8		
		MCDB	35.8	37.1	41.8	55.2	67.8	72.2	76.6	72.9	63.4	52.7	45.4	40.4		
	5%	WB	33.8	34.0	36.0	45.0	55.4	61.4	66.0	63.5	57.4	49.9	42.3	37.2		
		MCDB	34.2	34.6	38.8	52.1	65.6	69.7	74.0	70.4	61.3	51.1	43.1	37.7		
	10%	WB	32.3	32.3	34.0	41.9	52.3	59.0	64.0	61.7	55.5	47.7	40.2	35.2		
		MCDB	32.7	32.8	36.0	48.0	61.0	67.5	71.5	68.2	58.6	48.8	40.8	35.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	12.7	15.8	17.0	19.8	18.1	17.8	17.9	15.3	9.7	7.3	9.0		
		MCDBR	7.9	10.4	18.1	25.7	28.5	25.2	23.6	23.7	20.2	11.7	8.7	7.1		
		MCWBR	7.8	9.6	13.0	15.4	14.3	11.4	10.0	11.5	13.1	9.5	8.3	7.0		
		MCDBR	7.4	9.9	14.0	23.0	24.1	20.4	19.9	19.7	17.3	10.3	8.3	7.0		
	5% WB	MCWBR	7.4	9.4	10.7	14.5	13.3	10.5	9.9	10.7	12.0	9.1	8.0	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.234	0.285	0.312	0.353	0.350	0.347	0.371	0.356	0.332	0.312	0.254	0.356		
	taud		2.042	2.145	2.212	2.336	2.447	2.478	2.427	2.460	2.517	2.415	2.140	2.636		
	Ebn at Noon		135	208	250	264	276	278	267	262	247	204	127	84		
	Edh at Noon		11	21	29	32	31	31	31	28	22	17	10	6		
(44) All-Sky Solar Radiation	RadAvg		62	238	710	1155	1537	1652	1576	1217	699	282	73	30		
	RadStd		10	57	83	136	171	128	162	138	86	54	13	3		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+1.04	N/A	N/A	N/A	N/A	N/A	-346	-380	N/A
(47) Regional (45 neighbors)			+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page